

The completed presentation of $\pi_1(V)$ at the Lidman–Piccirillo cell $(m, n) = (0, 0)$

One line per relator; certificate pointers refer to the computation paper (*Simple connectivity of the Lidman–Piccirillo piece*, July 15, 2026 version) and the ancillary files. Harness: `decide.g/decide2.g`.

Generators. Fiber x, y, r, s (octagon edge loops of the genus-2 fiber F , based at p ; y is the LP curve class b , s the class d); base $A = \bar{\alpha}^{-1}$, $B = \bar{\beta}^{-1}$ (the based base loops through $q = (1/4, 1/4)$ of the cut square); meridians M (of T_α , based along $y_1 =$ initial segment of y to its c -crossing c_y), N (of T_β , based along $s_2 =$ initial segment of s to its e -crossing s_e). Monodromies: $\tilde{\varphi} =$ the handle swap, $\psi = h * \text{id}$, $h = T_a T_b: x \mapsto y^{-1}, y \mapsto yx$. Signs $\varepsilon_3, \varepsilon_4, \varepsilon_5, \varepsilon_A, \varepsilon_B \in \{\pm 1\}$ are swept convention parameters; the verdict $\pi_1 = 1$ holds for every assignment.

#	relator	certificate
R0	$[x, y] [r, s]$	genus-2 octagon surface relator (paper §2.1)
A1	$A x A^{-1} = r$	clean $\tilde{\varphi}$ -monodromy (swap); machine-certified lift, <code>model_check3.g</code>
A2	$A y A^{-1} = s$	same
A3	$A r A^{-1} = x$	same
B1	$B x B^{-1} = y^{-1}$	clean $\tilde{\psi}$ -monodromy ($h: x \mapsto y^{-1}$); <code>monodromy_check2.g</code>
B2	$B r B^{-1} = r$	clean ψ -monodromy ($h * \text{id}$ fixes right handle)
R3	$B (s^{-1} r^{-1} y x) B^{-1} = r^{-1} s^{-1} x$	completion relation $B \kappa_3 B^{-1} = \tilde{\psi}(\kappa_3)$, $\kappa_3 =$ third basis element of $\pi_1(F \setminus \nu c)$; engine record D5; paper §6
C1	$A s A^{-1} = N^{\varepsilon_3} y$	membrane $s \times \bar{\alpha}$ crosses T_β once, at $(s_e, \beta\text{-cut})$: $\bar{\alpha} \cdot (\beta\text{-cut}) = 1$, $s \cap e = \{s_e\}$ (position table D4); N 's s_2 -basing absorbs the lasso; paper §2.3 (worked membrane) + §4.3
C2	$B y B^{-1} = M^{\varepsilon_4} (yx)$	membrane $y \times \bar{\beta}$ crosses T_α once, at $(c_y, \alpha\text{-cut wrap})$: $\bar{\beta} \cdot (\alpha\text{-cut}) = 1$, $y \cap c = \{c_y\}$; M 's y_1 -basing absorbs the lasso; paper §2.3, Remark (worked membrane)
C3	$B s B^{-1} = \delta M^{\varepsilon_5} \delta^{-1} s, \quad \delta = r^{-1}$	membrane $s \times \bar{\beta}$ crosses T_α once, at $z_0 = (P_{s'}, \text{wrap})$, $s \cap c = \{P_{s'}\}$; lasso $\delta =$ slide of the meridian basing along c from $P_{s'}$ to c_y (engine D3; V4 arc-identity check); paper §4.3
F1	$M (A r^{-1})^{\varepsilon_A} A = 1$	T_α filling, direction $\alpha' c^n$, $n = 0$: $\text{dir}_{T_\alpha}^{\text{base}} = A r^{-1}$ (half-rotation drift = the lasso; $\varphi_0(c_y) = c_s$, $\tilde{\varphi}(y_1) = s_1$; framing lemma, paper §4.7); drift identity $2\lambda = 2\alpha' + [c]$ checked exactly
F2	$N ((r^{-1} M^{-\varepsilon_5} r) B)^{\varepsilon_B} = 1$	T_β filling, direction $\beta' e^m$, $m = 0$: $\text{dir}_{T_\beta}^{\text{base}} = (\delta M^{-\varepsilon_5} \delta^{-1}) B$ — the <i>pushoff-basing correction</i> (paper §4.5): the s_2 -whisker's transport membrane $s_2 \times \bar{\beta}$ crosses T_α once at the same z_0 (the c -crossing precedes the e -crossing on the s -edge, so $P_{s'} \in s_2$); meridian sign anti-coupled to ε_5 ; found by the T^4 known-answer calibration (paper §4.6)

Not present at $(0, 0)$ (enter only for $m, n \neq 0$): $\text{dir}_{T_\alpha}^{\text{fib}} = (rx)^{-1}$ (= the derived word of c), $\text{dir}_{T_\beta}^{\text{fib}} = s r^{-1} s^{-1}$ (= the s -whiskered word of e).

Verdict at $(0, 0)$. $H_1 = 0$ (all conventions); coset enumeration gives $|G| = 1$ in all 32 sign assignments of the completed presentation; Knuth–Bendix certifies every generator $\rightarrow 1$ in all conventions and in both based diagrams (`kb_certify_out2.txt`, `kb_diag2_full_out.txt`). Dropping F1–F2 gives $H_1 = \mathbb{Z}^2$; dropping one gives $H_1 = \mathbb{Z}$ (the homology bookkeeping of the reduction paper, Lemma 3.1).

Diff protocol. For each LP relator: match against C1–C3/F1–F2 modulo (i) global inversion of any generator, (ii) the swept signs, (iii) basing conjugation (whisker changes), and (iv) the V4 arc identity $\delta' = x$ vs. $\delta = r^{-1}$ ($\delta' \delta^{-1} \sim c$, immaterial in the group). A mismatch that survives (i)–(iv) localizes the disagreement to one geometric datum.